Classical Methods for Fabric Defect Detection Using the AITEX Dataset

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***Abstract*—Fabric defect detection is an important task in the textile industry because even small defects can affect the quality and appearance of the final product. In this week's work, we focused on implementing classical image processing techniques for identifying defects in fabric images.**

The AITEX Fabric Image Database was selected for this work. The dataset contains normal fabric images, defective fabric images, and corresponding defect masks. Two different approaches were implemented: Gabor filter-based texture analysis and local texture deviation.

The images were preprocessed, divided into training and testing sets, and used to develop reference texture maps. Threshold values were selected using the training data. Finally, the methods were evaluated using pixel-level precision, recall, F1-score, Intersection over Union (IoU), and image-level AUROC.

This work provides a basic comparison of two classical approaches for fabric defect detection without using deep learning models.

***Index Terms*—Fabric defect detection, AITEX dataset, image processing, Gabor filter, texture analysis, anomaly detection, computer vision.**

I. Introduction

Quality inspection plays an important role in textile manufacturing. During the production of fabrics, different types of defects such as holes, stains, broken threads, and irregular patterns may occur.

Traditionally, fabric inspection is performed manually. However, manual inspection requires time and may not always provide consistent results. Therefore, image processing techniques can be used to identify unusual patterns and possible defects in fabric images.

In this week's work, we explored classical image processing methods for detecting defects in fabric images. Instead of using a deep learning model, we focused on texture features and compared two approaches.

The first approach uses Gabor filters to capture texture information at different orientations. The second approach uses local texture deviation to identify regions whose texture differs from the normal fabric pattern.

II. Objectives

The main objectives of this week's work were:

- To understand and use the AITEX fabric image dataset.
- To separate normal and defective fabric images.
- To identify and pair defective images with their corresponding ground-truth masks.
- To preprocess images into a common size and grayscale format.
- To implement Gabor filter-based texture analysis.
- To implement a local texture deviation method.
- To select suitable thresholds using training data.
- To evaluate both approaches using standard performance metrics.

III. Dataset Description

For this work, we used the AITEX Fabric Image Database (AFID), which is publicly available from the AITEX website.

Dataset link: https://www.aitex.es/afid/

The dataset files used in this project are:

- `defect_images.zip`
- `nodefect_images.zip`
- `mask_images.zip`

The three archives contain defective fabric images, defect-free fabric images, and masks representing the defective regions.

The masks are important because they provide the ground-truth information required to compare the predicted defect regions with the actual defect regions.

A. Dataset Loading

The three dataset archives were uploaded manually because the source files were provided as compressed archives. The code then extracted the PNG images from the archives.

The ZIP files were extracted using Python's `zipfile` library. After extraction, all PNG images were collected from the dataset folders.

The image filenames were also checked to identify normal and defective images.

Images with the second filename field equal to `000` were treated as normal images. For defective images, the code searched for the corresponding mask using the image filename followed by `_mask.png`.

B. Dataset Statistics

After loading the images, the code displayed the number of normal images, defective image-mask pairs, and images with missing masks.

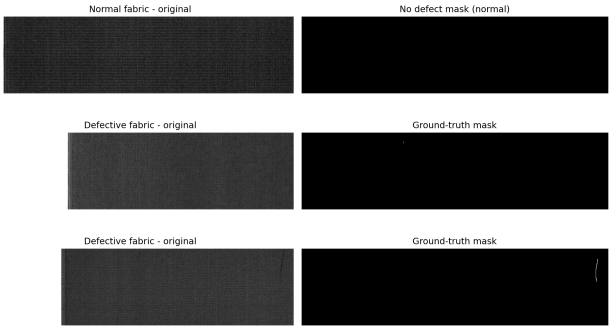


Fig. 1. Examples of normal and defective fabric images with their masks.

The processed dataset contained 140 normal images and 105 defective images with matched masks. One additional defective image did not have a matching mask and was not included in the usable image set. Thus, 245 images were used for the subsequent processing pipeline.

TABLE I
DATASET SUMMARY

Category	Number of Images
Normal images	140
Defective images with masks	105
Missing masks	1
Total usable images	245

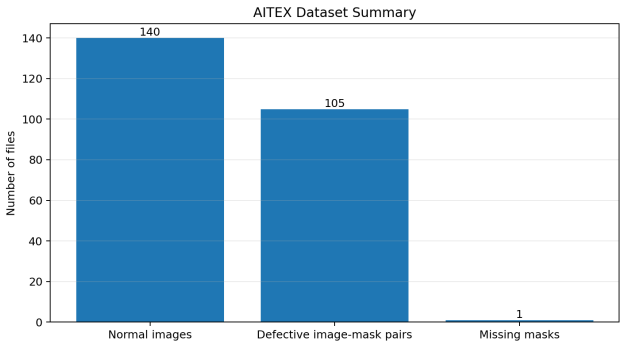


Fig. 2. Output showing the dataset statistics.

IV. IMAGE PREPROCESSING

Before applying the defect detection methods, all images were converted into grayscale and resized to a common resolution.

The selected image dimensions were:

- Width: 1024 pixels
- Height: 256 pixels

The images were normalized to the range of 0 to 1 by dividing their pixel values by 255.

The ground-truth masks were resized using nearest-neighbor interpolation to preserve the mask labels. A threshold of 127 was applied to convert the masks into binary images.

This preprocessing step ensured that all images had the same dimensions and could be processed consistently.

V. TRAINING AND TESTING SPLIT

After preprocessing, the normal images and defective image-mask pairs were combined into a single dataset.

Each image was assigned a label:

- 0: Normal fabric image
- 1: Defective fabric image

The dataset was divided into training and testing sets using the `train_test_split()` function from Scikit-learn.

We used the following settings:

- Training data: 75%
- Testing data: 25%
- Random state: 42
- Stratification: Based on image labels

TABLE II
TRAINING AND TESTING DATA DISTRIBUTION

Dataset	Number of Images
Total usable images	245
Training images (75%)	183
Testing images (25%)	62

The training set was used to create reference texture maps and select thresholds. The testing set was kept separate for final evaluation.

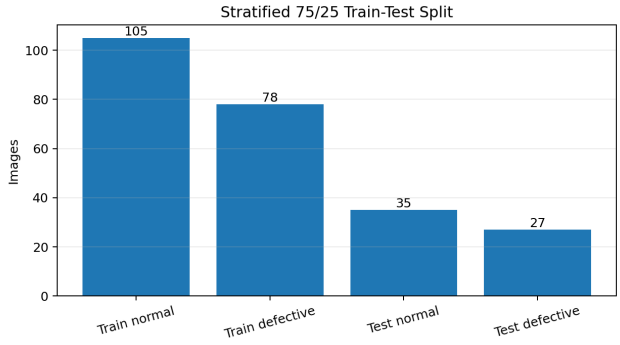


Fig. 3. Training and testing dataset distribution.

VI. GABOR FILTER-BASED DEFECT DETECTION

The first method implemented in this work uses Gabor filters for texture analysis.

Gabor filters are commonly used in image processing because they can capture texture patterns and directional information.

In our implementation, four different orientations were considered:

$$\theta \in \left\{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\right\} \tag{1}$$

A Gabor kernel of size 21×21 was used. The standard deviation was set to 4, wavelength to 10, and aspect ratio to 0.5.

Each filter was applied to the grayscale image. The absolute responses were calculated and averaged to obtain a combined Gabor response map.

A. Reference Gabor Map

The reference Gabor map was calculated by averaging the Gabor responses of normal images from the training set.

This reference represents the typical texture pattern of defect-free fabric.

For a test image, its Gabor response was compared with the reference map. The absolute difference was used as the anomaly score.

$$S_G(x, y) = |G(I)(x, y) - G_{\text{ref}}(x, y)| \tag{2}$$

Here, $G(I)$ represents the combined Gabor response of the input image and G_{ref} represents the reference Gabor map.

A higher score indicates a greater difference from the normal texture pattern.

VII. Local Texture Deviation

The second method implemented was local texture deviation. This method identifies unusual regions by measuring local intensity variation.

For each image, the local mean and local mean of squared intensity values were calculated using a 15×15 averaging window.

The local variance was calculated as:

$$V(x, y) = \mu_2(x, y) - \mu(x, y)^2 \tag{3}$$

Where:

- $\mu(x, y)$ is the local mean intensity.
- $\mu_2(x, y)$ is the local mean of squared intensity values.

The local standard deviation was then calculated as:

$$T(I)(x, y) = \sqrt{\max(V(x, y), 0)} \tag{4}$$

The reference local texture map was obtained by averaging the local texture maps of normal training images.

The anomaly score was calculated by comparing the input image's local texture map with the reference:

$$S_T(x, y) = |T(I)(x, y) - T_{\text{ref}}(x, y)| \tag{5}$$

This method helps identify areas where the local texture differs from the normal fabric pattern.

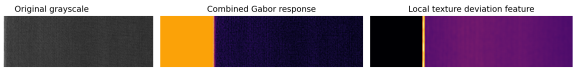


Fig. 4. Texture feature maps produced by the two methods.

VIII. Threshold Selection

After generating anomaly score maps, a threshold was required to convert them into binary defect masks.

Pixels with anomaly scores greater than or equal to the selected threshold were considered defective.

$$P(x, y) = \begin{cases} 1, & S(x, y) \geq \tau \\ 0, & S(x, y) < \tau \end{cases} \tag{6}$$

Here, τ represents the threshold value.

The code tested 37 threshold values between 0.05 and 0.95.

For each method, the F1-score was calculated on the training images for each threshold. The threshold that produced the highest average training F1-score was selected.

Separate thresholds were selected for the Gabor and local texture methods.

TABLE III
Selected Threshold Values

Method	Threshold
Gabor	0.0500
Local Texture	0.1250

Threshold selection (training set)

Gabor threshold: 0.050

Local texture threshold: 0.125

Thresholds selected using a 37-value search; F1 computed on downsampled training maps for speed.

Fig. 5. Training-selected threshold values.

IX. Performance Evaluation

Both methods were evaluated on the held-out testing images.

The following performance metrics were calculated.

A. Precision

Precision measures how many pixels predicted as defective were actually defective.

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

B. Recall

Recall measures how many actual defective pixels were correctly identified.

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

C. F1-Score

The F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{9}$$

D. Intersection over Union

IoU measures the overlap between the predicted defect mask and the ground-truth mask.

$$IoU = \frac{TP}{TP + FP + FN} \tag{10}$$

E. Area Under the ROC Curve

Image-level AUROC was calculated using the maximum anomaly score of each testing image.

It measures how well the maximum anomaly score distinguishes normal images from defective images.

All pixel-level metrics were calculated by comparing the predicted masks with their corresponding ground-truth masks.

X. Results and Observations

The notebook generates individual testing-image metrics and an overall summary for both methods.

The results include pixel-level precision, recall, F1-score, IoU, and image-level AUROC.

The numerical values below are taken from the generated held-out summary CSV file. The very low pixel-level precision, F1-score, and IoU indicate that the current thresholded predictions do not localize the defect pixels well. The AUROC values of 0.5 indicate chance-level image-level discrimination in this evaluation.

TABLE IV
Performance Comparison of Both Methods

Metric	Gabor	Local Texture
Pixel Precision	0.0001	0.0002
Pixel Recall	0.3673	0.1767
Pixel F1-score	0.0002	0.0004
Pixel IoU	0.0001	0.0002
Image-level AUROC	0.5000	0.5000

Held-out Test Set Performance Summary

method	pixel_precision	pixel_recall	pixel_f1	pixel_iou	image_level_AUROC
gabor	0.0001	0.3673	0.0002	0.0001	0.5
local_texture	0.0002	0.1767	0.0004	0.0002	0.5

Fig. 6. Performance summary generated from held-out test results.

A. Visual Results

For visual evaluation, the notebook generates examples containing four different views:

- Original grayscale fabric image.
- Ground-truth defect mask.
- Anomaly score map.
- Predicted defect mask overlaid on the original image.

These visualizations help us understand how each method identifies unusual fabric regions.

B. Observations

The two approaches use different ways of analysing fabric texture.

The Gabor method focuses on directional texture patterns, while local texture deviation focuses on changes in local intensity variation.

The comparison of their numerical results and visualizations will help us understand which method is more suitable for the selected dataset.

The performance may vary depending on the type, size, and appearance of the fabric defect.

XI. Generated Output Files

The notebook also saves the evaluation results for further analysis.

The following output files are generated:

- `heldout_per_image_metrics.csv`
- `heldout_summary.csv`
- `06_gabor_detection_results.png`
- `06_local_texture_detection_results.png`
- `07_performance_summary.png`
- `08_generated_outputs.png`
- `heldout_summary.csv`
- `heldout_per_image_metrics.csv`

These files make it easier to compare the methods, inspect individual predictions, and use the results in future work.

XII. Challenges Faced

During this week's implementation, we worked on several practical aspects of image processing.

Some of the main challenges were:

- Downloading and extracting the dataset correctly.
- Handling the three separate ZIP archives.
- Matching defective images with their corresponding masks.

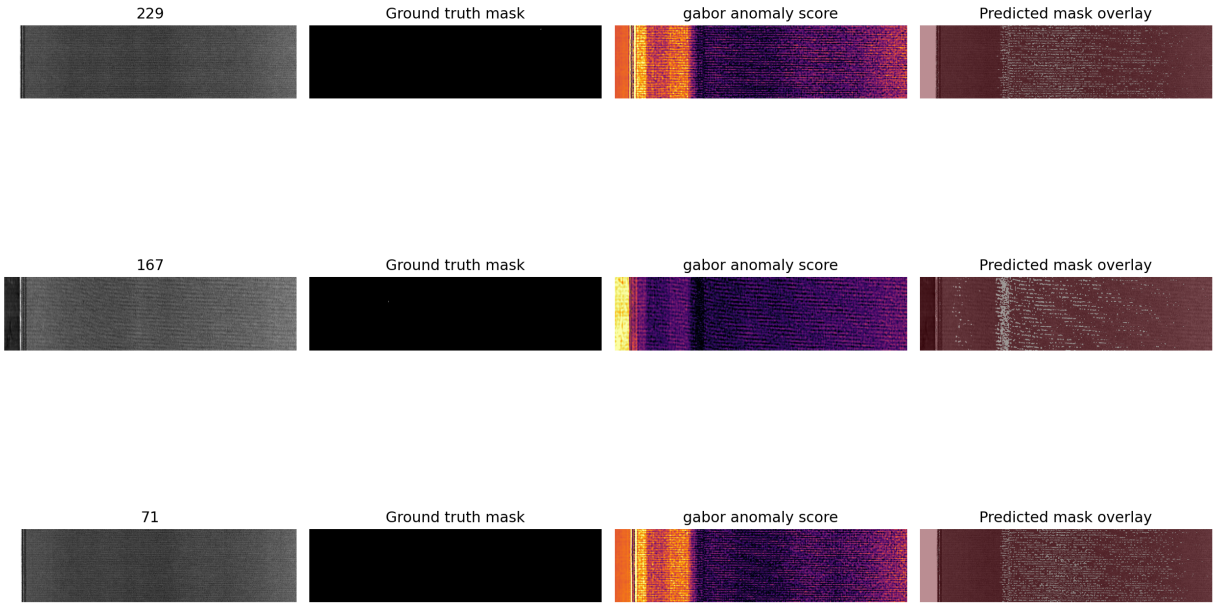


Fig. 7. Visual defect detection results using the Gabor method.

- Making sure all images and masks had consistent dimensions.
- Selecting suitable threshold values for both detection methods.
- Evaluating predicted defect regions using ground-truth masks.

We also simplified the dataset loading stage to avoid unnecessary network waiting and repeated extraction.

XIII. Conclusion

In this week's work, we implemented two classical image processing approaches for fabric defect detection using the AITEX dataset.

The first approach used Gabor filters to analyse directional texture patterns, while the second approach used local texture deviation to identify unusual regions.

We also completed the dataset preparation, preprocessing, training and testing split, threshold selection, and evaluation pipeline.

The use of ground-truth masks allowed us to compare the predicted defect regions with the actual defect regions.

Overall, this work helped us understand how traditional image processing techniques can be used for fabric inspection and how different texture features can help identify defects.

XIV. Future Work

In the next stage, we can improve this work by exploring additional texture features and testing different threshold selection strategies.

We can also investigate combining both methods to improve defect localization.

Finally, the results obtained from these classical methods can be used as a baseline for comparison with machine learning and deep learning approaches.

References

[1] AITEX, "AITEX Fabric Image Database (AFID)," Available: https://www.aitex.es/afid/.
[2] OpenCV, "Open Source Computer Vision Library," Available: https://opencv.org/.
[3] Scikit-learn, "Machine Learning in Python," Available: https://scikit-learn.org/.

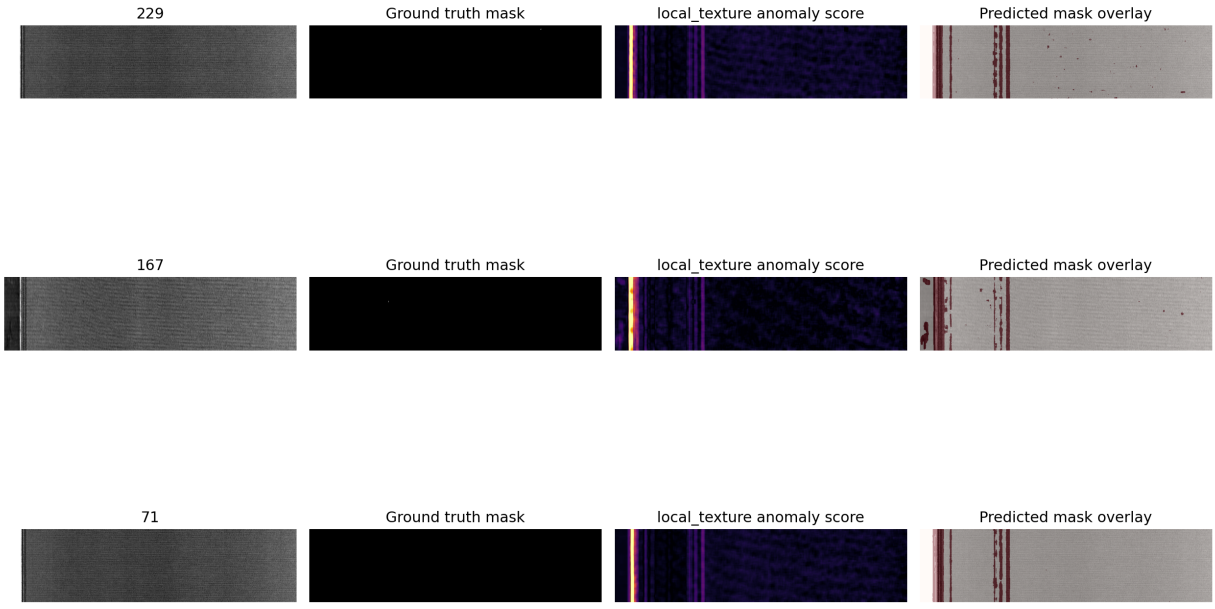


Fig. 8. Visual defect detection results using local texture deviation.

Generated report outputs

- ✓ heldout_per_image_metrics.csv
- ✓ heldout_summary.csv
- ✓ Gabor example visualizations
- ✓ Local texture example visualizations
- ✓ week7_results_summary.txt
- ✓ week7_outputs.zip

Fig. 9. Generated evaluation and visualization files.